



SWE4600:Final Year Project Step by Step Guide



Presented by :

Maryam Ibrahim Mukhtar

Department of Software Engineering

Bayero University Kano.

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Outline

- Introduction
- General Formatting
- Preliminary Pages
- Chapter One
- Chapter Two
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- Assessment
- Important Deadlines
- Conclusion



Introduction

- In most universities all over the world, undergraduate students in their final year are expected to complete a research project as part of their requirements for graduation.
- Students going through Software Engineering or related program are normally required to execute software projects which involve going through the complete system development cycle from beginning to end.
- In spite of the fact that such students take courses on research methodology and Software Engineering during their programs, many of them, by the time they start these projects, have little or no ideas on how to systematically carry out such software projects.
- This presentation is to guide such students in how to carry out Software Engineering projects from beginning to end and how to fully prepare the documentation for these projects.



General Formatting

- The writing should be in Times New Roman Font Type. 12points, Line Spacing 1.5
- Chapters Heading should be in Centralized Alignment
- Sub-Sections should be in Left Alignment
 - Inside each Subsection, the writing Should be Justified aligned.
- APA reference style should be used.
- Page Numbering should be in center of a page.
- Each Chapter should start on a new page.
- All abbreviations must be written completely the first time they are used with parenthesis around the abbreviation.
- All figures and tables must be numbered according to the chapter and referenced inside the chapter.
 - For example the first figure in chapter 3 should be labelled Figure 3.1, the fourth table in chapter 4 should be Table 4.4.
 - Tables are labelled at the **Top**, Figures are labelled at **the bottom**.
 - Both Tables and Figures and their labels are center aligned.



Preliminary Pages

- Preliminary pages include several key sections that provide essential information and context for the main content.
- The preliminary pages of **SWE4600:Final Year Project** should contain the following:
 - Title Page
 - Declaration
 - Certification
 - Dedication
 - Acknowledgement
 - Table of Content
 - List of Tables
 - List of Figures
 - Abstract
- Preliminary pages are numbered in **Roman Numerals**
 - Title page has no page numbering



Title Page: The Format

Title of the Project

BY

Name

Registration Number

A PROJECT SUBMITTED TO THE DEPARTMENT OF SOFTWARE
ENGINEERING, FACULTY OF COMPUTING, BAYERO UNIVERSITY
KANO, IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE
AWARD OF BACHELOR OF SCIENCE (B.Sc.) DEGREE IN SOFTWARE
ENGINEERING

Month, Year



Declaration Page: The Format

DECLARATION

I hereby declare that the project titled " ~~XXXXXXXXXXXX~~ " submitted to **Department of Software Engineering, Bayero University, Kano**, is a record of an original work done by me under the guidance of **Name of the Supervisor**. This project has not been submitted for the award of any other degree or diploma, either in full or in part, to any other institution or university.

Signature & Date

Full Name
Registration Number



Certification Page: The Format

CERTIFICATION

This is to certify that the project titled "~~XXXXXXXXXX~~" is carried out by **Full Name** with **Registration number**, a student of **Software Engineering Department**, Faculty of Computing, Bayero University Kano. This project has been submitted in partial fulfillment of the requirements for the award of BACHELOR OF SCIENCE (B.Sc.) degree in Software Engineering.

Name

Project Supervisor

Signature & Date

Name

Project Coordinator

Signature & Date

Name

Head of Department

Signature & Date

Name

External Examiner

Signature & Date



Acknowledgement, Table of Content, List of Tables and Figures, Abstract

- **Acknowledgements:** A section where the author(s) can thank individuals, organizations, or institutions that helped in the completion of the project.
- **Table of Contents:** Lists the main sections and subsections of the project, along with their page numbers, providing an organized outline of the document.
- **List of Figures and Tables:** If the project includes various figures, tables, or illustrations, this list helps readers quickly locate these items.
- **Abstract:** A brief summary of the project, including: Background(one sentence),a clear statement of the aim, methodology(a sentence on each chapter)and conclusions.
 - **Written in present tense.**



Chapter One

Introduction

- This Chapter of **SWE4600:Final Year Project** should contain the following:

- 1.1 Background of Study
- 1.2 Problem Statement
- 1.3 Aim and Objectives
- 1.4 Significance of the Study
- 1.5 Scope
- 1.6 Summary



Background of Study

- Background of Study: Here students are to provide an overview of the topic, summarize existing knowledge, and explain the problem the research will address. You must use citation
- Example: Background of Study for Student Record System

In today's educational landscape, many institutions still rely on outdated or manual systems for managing student records. These traditional methods often involve cumbersome paper-based processes or inefficient legacy software that can lead to significant challenges. Smith and Jones (2018) note that such methods are increasingly problematic due to their inherent limitations, including frequent errors, slow processing times, and difficulty in maintaining up-to-date information. For instance, manual data entry is prone to human error, which can compromise the accuracy and reliability of student records.

The need for efficiency and automation in educational record management is further highlighted by Ali et al. (2020). Their research shows that automation can greatly enhance the accuracy of data management and reduce the administrative burden on staff. Automated systems can streamline processes such as data entry, retrieval, and reporting, which not only improves operational efficiency but also reduces the likelihood of errors.

In response to these issues, this study aims to develop a modern Student Record System that addresses the shortcomings of current systems. The proposed system will incorporate advanced features such as automated data management, enhanced security protocols, and user-friendly interfaces. By automating record-keeping processes and integrating secure access controls, the new system is expected to improve the accuracy of student data, enhance data protection, and streamline administrative tasks. This approach aligns with the growing need for more efficient and secure record management solutions in educational institutions (Ali et al., 2020; Garcia, 2019).

In summary, the background of this study highlights the significant challenges associated with traditional student record systems and presents the development of a modern system as a solution. By addressing issues related to accuracy, security, and efficiency, this research aims to contribute to more effective management of student records and provide valuable support for educational institutions.



Problem Statement

- Problem Statement is a clear description of the problem that the project is addressing.
- The problem statement should generally have two (2) paragraphs. The first paragraph should state the problem that is the reason for the project. The second paragraph should describe the proposed solution.
- Example: Problem Statement for a Student Record System

The current student record management system at XYZ university involves extensive paperwork for managing student enrolment, academic records, and attendance. This process is error-prone, time-consuming, and prone to data loss or mismanagement. Additionally, generating reports and retrieving student information requires considerable effort and often results in delays.

Hence the need for an automated system that solves the above mentioned problems



Aim and Objectives

- The aim should normally be written using an infinitive verb, something like: The aim of this project is to develop, to generate, to map, to build, to investigate, to explore
- Objectives are often presented as a formatted list and they should clearly make visible the sequence of **big steps** in the project with each step having a definable outcome.

Example

The Aim of this project is to develop an event hall booking system that simplifies the process of reserving and managing event spaces. The Objectives are:

- i. To design and implement functionalities for searching, viewing, and reserving event halls.
- ii. To design and implement set up notifications feature to confirm bookings and remind users of upcoming events.
- iii. To design and implement an admin panel for managing bookings and updating hall information.



Significance of Study

- Significance of the Study: Here students are to explain the importance and potential impact of the research.
- It should outlines how the project contributes to existing knowledge, solves a problem, or benefits specific groups.
- Example: Significance of Study for a Student Record System

This project aims to improve how educational institutions manage student records by developing an efficient, automated system. The system will reduce manual errors, enhance data security, and allow easy access to student information, benefiting both administrators and students. It will also save time and resources, contributing to a smoother administrative process and better decision-making based on accurate data. This study can serve as a foundation for future improvements in student record management systems.



Scope

- Scope: Here students are to define the boundaries and specific aspects that their study will cover.
- It should outlines what the project will include, what it will not address, and any limitations or constraints.
- Example: Scope for a Student Record System

This project will develop a Student Record System that automates student data management. The system will include:

1.Features: Automated data entry, secure login, real-time updates, and simple interfaces for users.

2.Users: Administrative staff and students.

The project will not include mobile app development or integration with external systems. It will focus only on creating a web-based system for managing student records.



Summary

- **Summary:** Here students are to to reinforce the chapter's content and ensure that readers understand the essential takeaways.

- **Example :** Chapter 1 summary for a topic Student Record System

This Chapter introduced the development of a Student Record System designed to enhance the management of student data in educational institutions. The chapter outlined the challenges associated with traditional record-keeping methods, including inefficiencies and security concerns. Key points discussed include the need for improved data accuracy, automated processes, and secure access controls. The chapter also highlighted the objectives of the project, which include creating a system with automated data entry, real-time updates, and user-friendly interfaces for administrative staff and students. The chapter set the stage for the subsequent sections by establishing the context and importance of the project, paving the way for a detailed exploration of the system's design and implementation in the following chapters.



Chapter Two

Literature Review

- This Chapter of **SWE4600:Final Year Project** should contain the following:

2.1 Introduction

2.2 Overview of Concept (Optional subsection)

2.2 Related Works

2.3 Summary



Introduction, Related Work, Summary

- Introduction : Here students are to briefly explain what the chapter will contain.
- Related Works
 - Students are to review at least 10 papers related to their work.
 - Since the project is on Development of software, the review should be on projects focused on developing Software.
 - Papers included in your review should be published not later than 10 years ago (2014-Date).
 - Papers must be arranged in Ascending or Descending order.
 - Use Google Scholar database
 - Tip: Use a table to create the review, then translate the table—**See slide 19 for an example.**
- Summary : Here students are to provide summary of all the things covered in the chapter.



Literature Review Table

Table 2.1: Summary of Related works

Author(s)	Year	Title	Research Focus	Programming Languages & Software Type (Methodology)	Key Functionalities Identified
Kumar & Patel	2022	User Interface Design for Educational Software	Explored user-friendly design in educational software	JavaScript, React (Web-based software)	Intuitive dashboard, mobile compatibility, notifications
Wang & Lee	2021	Security Concerns in Student Information Systems	Focused on data security in student record systems	Python, Django (Web-based software)	Data encryption, user authentication, audit trails
Ali et al.	2020	Automation in Student Data Systems	Investigated automation in student record systems	PHP, MySQL (Web-based software)	Automated data backup, attendance tracking, performance analysis

Sample translation

Ali et al. (2020) investigated automation in student record systems. Their software is web application developed using PHP programming language and MYSQL database. Their application provides functionalities such as automated data backup, attendance tracking and performance analysis.



Chapter Three

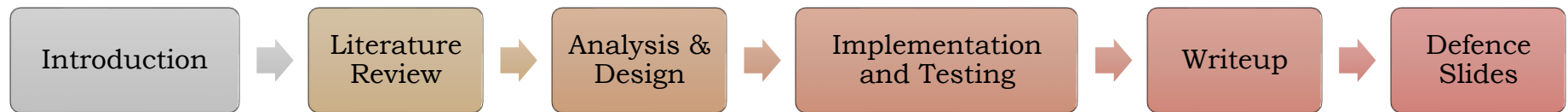
Methodology

- This Chapter of **SWE4600:Final Year Project** should contain the following:
 - 3.1 Introduction
 - 3.2 Project Workflow
 - 3.3 System Development Model
 - 3.4 Analysis of Existing and Proposed System
 - 3.4.1 Description of Existing System
 - 3.4.2 Requirement Elicitation
 - 3.4.3 Requirements Definition
 - 3.4.4 Requirement Analysis
 - 3.5 System Design
 - 3.5.1 Description of Proposed System
 - 3.5.2 Architecture Design
 - 3.5.3 Database Design
 - 3.6. Summary



Project Workflow, SDLC, Existing System

- Project Workflow: Here students are to provide a diagram showing the various stages their project will follow from beginning to end.
- Example:



- System Development Model : Here students are to choose an SDLC for their project and briefly explain why it is selected.
- Description of Existing System: Here students are to describe **in details** how the existing system works, who interact with it and what is his/her role/function. **See slide 22 for an example**



Description Existing System

- Example for Student Record System.

- **Administrative Staff:**

- **Role:** Handle all student records, including enrolment forms, grades, attendance sheets, and personal details. They are responsible for data entry, updating records, and managing physical files.

- **Teachers/Faculty:**

- **Role:** Provide updates on student performance, attendance, and other academic information using paper forms or handwritten notes. They submit these records to administrative staff.

- **Students:**

- **Role:** Access their own records, grades, and attendance by requesting physical copies or information from administrative staff. They may also submit paper forms for updates or changes to their records.



Elicitation, Definition and Analysis

- Requirement Elicitation: Here students are to provide detailed explanation of their findings during requirements elicitation.
 - Evidence of elicitation must be attached in appendix **See slide 24 for an example.**
- Requirements Definition: Here students are to state the User requirements and non-functional requirements of their system. **See slide 25 for an example.**
- Requirement Analysis: Here students are to analyze their requirements using **Use Case diagram** and its description. **See slide 26 and 27 for an example.**



Requirement Elicitation

- Example for Student Record System.
- This section summarizes the findings from interviews conducted to gather requirements for the development of a Student Record System. The Stakeholders Interviewed include Administrative Staff , Teachers/Faculty and Students. Interview Findings:
- **Students**
 - **Key Needs:**
 - **Self-Service Portal:** Access to personal records, grades, and attendance.
 - **Update Requests:** Ability to request updates or corrections to their records.
 - **Data Security:** Assurance that personal data is protected and only accessible to authorized users.
 - **Challenges:**
 - Limited access to up-to-date personal records.
 - Dependence on administrative staff for record updates and information retrieval.



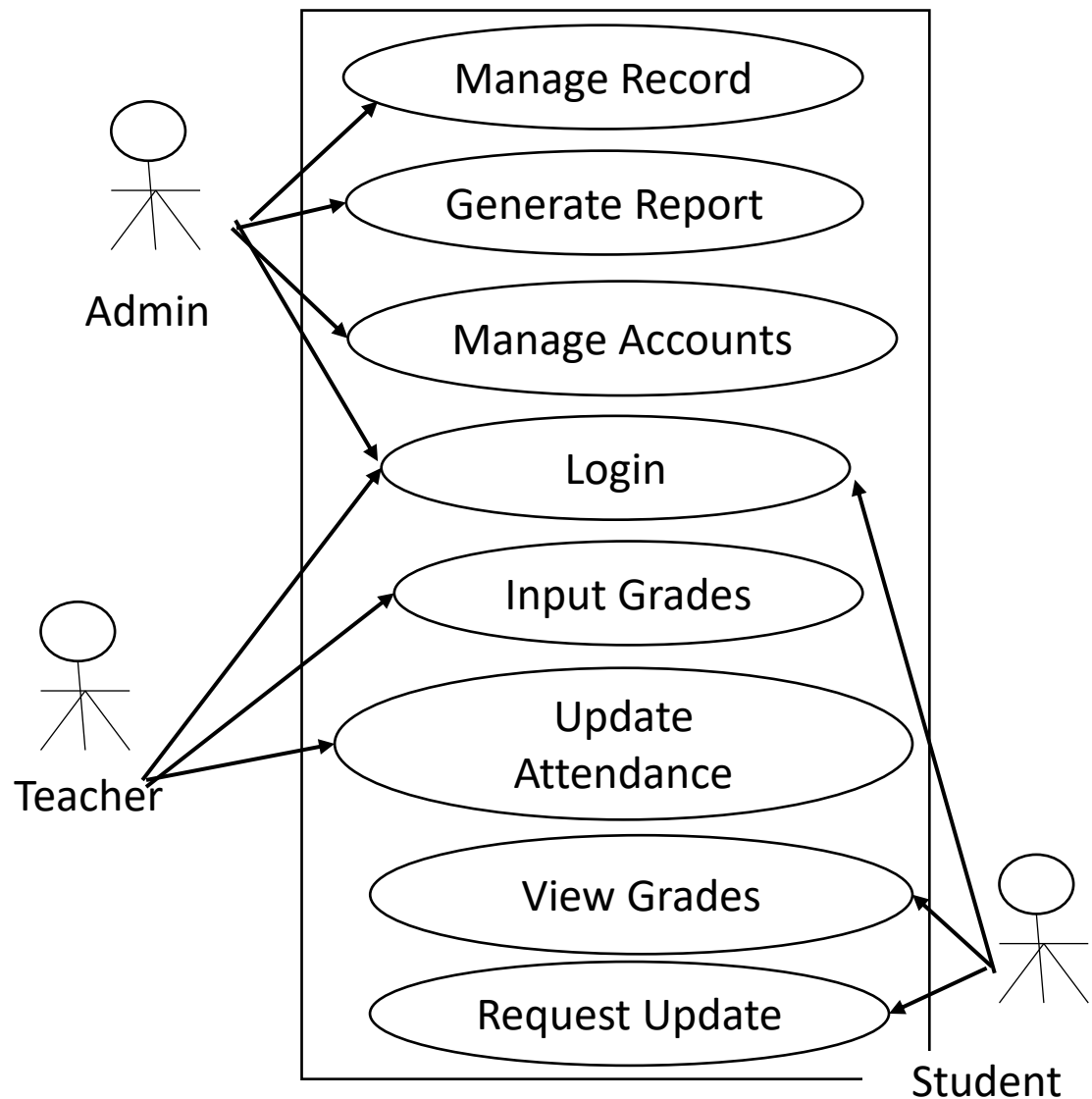
Requirement Definition

- Example for Student Record System.
- **User Requirements:**
 - Administrators shall be able to add, update, and delete student records.
 - Administrators should be able to generate reports.
 - Administrators shall be should be able Manage User Accounts.
 - Teachers should be able to input student grades and update and attendance.
 - Students shall be able to view their own grades and attendance records.
- **Non-Functional Requirements:**
 - *The system must have role-based access to ensure that only authorized personnel can view or modify sensitive information.
 - The system should handle high volumes of concurrent users and data transactions efficiently.
 - The system should be reliable and Include backup and recovery features to prevent data loss.



Requirement Analysis- Use Case Diagram

- Example for Student Record System.
- Actors:
 - Administrative Staff, Teachers, Students
- Use Cases:
 - Add/Update/Delete Student Record --- (Manage Record)
 - Generate Reports,
 - Manage User Accounts
 - Input Grades,
 - Update Attendance,
 - View grades and Attendance Records, Request Record Update
 - Login



Requirement Analysis- Use Case Description

- Example for Login Use Case -Student Record System.

Use Case Name : Login	Priority : High
Actor: Shopper	
Description: This use case documents the procedure for logging into the library management system.	
Trigger : External	
Preconditions : The system database is up-to-date and on-line	
Normal Course: 1) The system requests that the actor enters his/her user id and password. 2) The actors enters user id and password. 3) The system validates the user id and password (Alternative course). 4) The user is logged into the system 5) The use case ends.	
Alternative course: 4b. The user id or password is invalid 1) The System ask the user for a valid password 2) The User enters another wrong password 3) The system blocks the user 4) The Use Case ends	
Post Conditions: The login record is saved.	
Exceptions: 1) The System displays "Invalid User name or password" 2) The System displays "Re-enter a valid password" 3) The system display "You have been blocked contact admin to resolve login issue"	



System Design

- System Design: Here students are to briefly explain what the section will contain.
- Description of Proposed System: Here students are to describe their system using an activity and/or sequence and/or state diagram.
- Architecture Design: Here students are to describe their system using class diagram and/or Package Diagram
- Database Design: Here students are to describe their system using Entity Relationship (ER) diagram.



Chapter Four

Implementation & Testing

- This Chapter of **SWE4600:Final Year Project** should contain the following:

4.1 Introduction

4.2 Implementation

4.2.1 Implementation Tools

4.2.2 Algorithms of major functionality

4.2.3 Description of System Operation (using sample interfaces)

4.3 Testing

4.3.1 Unit testing

4.3.3 Integration testing

4.3.4 System testing

4.3.5 Usability Testing

4.4 Summary



Implementation

- Implementation: Here students are to briefly explain what the section will contain.
- Implementation Tools : Here students are to state the tools such as operating System, IDE, Programming Language, Database etc.. they used and briefly explain why they choose them.
- Algorithms of major functionality : Here students are to provide the algorithms used to create the major functionalities in their system.
- Description of System Operation: Here students are to provide screen captures of the developed system.
 - Students should capture them in such a way that the person reading the project can understand the way the system works.
 - It should capture the functionalities which algorithms were given in 4.2.2.



Testing

- Testing: Here students are to explain what the section will contain.
- Unit Testing: Here students are to provide unit test cases of their major functionalities. They should be align to the functional in 4.2.2
- Integration testing: Here students are to provide Integration test cases of their major functionalities.
- System testing: Here students are to explain how the whole system was tested to make sure everything is working.
- Usability Testing: Here students are to do a usability test of their developed system (Using a suitable questionnaire)



Chapter Five

Summary, Conclusion and Recommendation

- This Chapter of **SWE4600:Final Year Project** should contain the following:

- 5.1 Summary
- 5.2 Conclusion
- 5.3 Limitation
- 5.4 Recommendation



Summary, Conclusion, Limitation and Recommendation

- Summary: Here students are to write a paragraph each summarizing the content of all the chapters in their project.
- Conclusion: Here Students are to capture brief introduction to the project, problem statement, methodology used and key results obtained(Functionalities of their software)
- Limitation: Here Students are to state functionalities that should be in their system but are not.
 - Example an online ordering system that does not contain a real payment gateway.
- Recommendation: Here Students are to indicate areas (functionalities) to expand the project further.



Appendix

- References: Here students are to provide list of all the cited works.
 - Any work cited must reference.
 - Any reference must have been cited in the work.
- Elicitation evidence: Here Students are expected to provide evidence of the elicitation conducted in 3.4.2 (example a copy of questions used in eliciting the requirements)
- System Testing Evidence, Here Students are expected to provide evidence of the usability testing conducted in 4.3.5 (example a copy of questions used in eliciting the requirements)
- Codes of major functionality– Here Students are expected to provide section of codes aligning to 4.2.2 (algorithm for major functionalities)



Defense Slides

- The Defence Slides of **SWE4600:Final Year Project** should contain the following:
- Slide 1: Project Title, Student Name & ID, Supervisor Name
- Slide 2: Outline
- Slide 3 : Introduction
- Slide 4: Problem Statement
- Slide 5: Aim and objectives
- Slide 6 :A summary of literatures reviewed
- Slide 7 : Methodology Flow
- Slide 8 :Use Case Diagram
- Slide 9: Summary of Implementation Tools
- Slide 10: Summary of Testing stages
- Slide 11:Conclusions
- Slide 12: References



Assessment

- **SWE4600:Final Year Project** is assess as following:
 - Supervisor: 60 marks
 - Assessment area all the chapters and software
 - Internal and External Defense: 40 marks
 - Assessment areas include Composure, Quality of the slides, demonstrating the software, navigating through source code and database.



Important Deadlines

- Undergraduate Project Topic Submission and Approval – A Week after mid semester
- Chapter 1 and 2 Submission and Approval : A week to first semester Exams
- Chapter 3 Submission and Approval : First Week in Second Semester
- Chapter 4 and 5 Submission and Approval : 4th Week in Second Semester
- **Exact Dates and Time will be communicated to you.**



Conclusion

- This presentation is to guide students on how to write their final year project.
- It also guide students on what to include in defense presentations slides of their projects after they have completed them.



Questions??